

Athiyaman M

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PROFESSIONAL SUMMARY

Software Engineer with **1+ year of experience** building **production-scale backend systems**, AI-powered enterprise applications, and healthcare software. Currently working as a **Backend Software Engineer** at the Centre for Development of Advanced Computing (C-DAC), India's premier **R&D organization** under MeitY, developing mission-critical national examination infrastructure using **Python, FastAPI, PostgreSQL, Elasticsearch**, and distributed log analytics. Previously contributed to **clinical software** development at **HTIC, IIT Madras**. Experienced in backend, REST API design, distributed systems, AI/LLMs, RAG, database optimization, and scalable production software.

EXPERIENCE

Project Engineer - Backend Software Engineer

Aug 2025 – Present | Chennai, India.

Centre for Development of Advanced Computing (C-DAC), Government of India.

- Developed and maintained a large-scale examination **server-side security monitoring and risk-scoring system** deployed on **PXE** servers across government examination centers nationwide, enabling real-time monitoring of thousands of client systems during SSC live exams.
- Designed and architected **FastAPI**-based backend APIs for anomaly detection, device validation, security compliance monitoring, and automated risk assessment using **30+ security indicators**, **5+ data sources**, and severity-based scoring models, with real-time enforcement through API-driven blocking mechanisms.
- Engineered production-grade solutions leveraging **Python, MariaDB, SQLite, REST APIs**, Regex-based log parsing, **JSON** processing, structured **logging**, boot validation, and security telemetry analysis to ensure examination infrastructure integrity.
- Built large-scale **log analytics** and monitoring pipelines using **Elasticsearch, Kibana, Regex**, and structured logging frameworks, enabling centralized security monitoring, forensic analysis, and operational reporting across distributed examination environments.
- Delivered **production-grade**, mission-critical software supporting **thousands of client systems** across government examination centers nationwide, ensuring **secure, scalable, and tamper-resistant** examination infrastructure.
- Developed AI-powered **proof-of-concepts** for financial document intelligence, research paper summarization, and explainable **healthcare AI** applications using Large Language Models (**LLMs**), Retrieval-Augmented Generation (**RAG**), and modern AI frameworks.
- Contributed to the **BharatDB** initiative by optimizing SQL queries and PostgreSQL database operations, performing database performance analysis, and identifying, reporting, and **resolving 3 open-source PostgreSQL bugs**.

Project Intern - Healthcare Software Developer

Jun 2024 – Jul 2025 | Chennai, India.

Healthcare Technology Innovation Centre (HTIC), IIT Madras.

- Developed healthcare software applications for **Total Knee Arthroplasty (TKA)** planning using **QT, C++, VTK, and ITK**.
- Built medical **image-processing** workflows for CT-based bone segmentation, **3D reconstruction**, visualization, and surgical planning.
- Integrated REST APIs and **MySQL** databases for healthcare data management and clinical workflow automation.
- Utilized **Azure DevOps, Git, CMake, and CI/CD** practices for collaborative development, testing, version control, and deployment.

PROJECTS

[AuditShakti - AI Compliance & Audit Intelligence Platform](#), | *Python, FastAPI, Next.js, scikit-learn, PyMuPDF, Tesseract OCR, Gemini, SQLite*

- **Problem:** Developed an AI-powered **compliance and audit** platform to automate financial document **review, regulatory validation, risk assessment, and audit report** generation, reducing manual effort and improving compliance transparency.
- **Approach:** Built an end-to-end document intelligence pipeline with **PDF** parsing, **OCR** fallback, rule-based compliance validation (Ind-AS, Schedule III, SEBI, RBI, ESG), hybrid retrieval (SQLite FTS5 + semantic search), Google Gemini-powered evidence-based Q&A, and portfolio risk analytics using FastAPI and Next.js.
- **Result:** Delivered an **explainable decision-support system** with page-level evidence traceability, real-time compliance analytics, conversational AI assistance, and exportable JSON/Excel/PDF reports, enabling faster and more accurate audit workflows.

[Research Paper Intelligence using Multi-LLM RAG](#), | *Python, Streamlit, LangChain, Groq API, Ollama, Llama 3.3, Gemma, PyMuPDF*

- **Problem:** Developed an AI-powered **research assistant** to automate scientific paper summarization, key insight extraction, and multi-document **comparative analysis** while addressing structure preservation and factual consistency challenges in LLM-generated summaries.
- **Approach:** Built a **section-aware document intelligence** pipeline with PyMuPDF-based PDF parsing, heuristic section detection, multi-provider LLM orchestration (**Groq API, Ollama, local models**), fact-consistency auditing, domain-adaptive prompting, and a custom ROUGE-based evaluation framework.
- **Result:** Delivered an interactive Streamlit application for comparative literature analysis, survey synthesis, and publication-ready evaluation, improving **ROUGE-2 F1 by 98%**, **factual consistency by 55%**, and **section coverage by 33%** over baseline while eliminating factual contradictions in the final summaries.

[Explainable Mental Health Risk Prediction System](#), | *Python, PyTorch, Flask API, Scikit-learn, SHAP, LIME, Pandas, Streamlit*

- **Problem:** Built an early mental health risk detection system to **classify Low / Medium / High risk** from social media or journal text, addressing the core challenge of poor interpretability in black-box ML systems.
- **Approach:** Designed a **multimodal NLP + XAI** pipeline using Multilingual **BERT embeddings**, behavioral feature fusion, and **Logistic Regression**. Benchmarked **TF-IDF, SVM, and Random Forest** baselines - deployed the solution with a Flask API, Streamlit dashboard, SQLite feedback loop, and **SHAP / LIME / Counterfactual** Explanations.
- **Result:** Delivered an **explainable decision-support system** with real-time predictions, human-in-the-loop review, and monitoring, repo validation docs report **94% accuracy** with ~920 ms end-to-end latency, while the research benchmark reports Macro F1 0.6464 and Macro AUC-ROC 0.8726 for the best balanced model.

SKILLS

- **Programming** : Python, SQL, C++
- **Backend** : FastAPI, Flask, REST API Design, Backend Architecture, Microservices, ETL Pipelines
- **Databases** : PostgreSQL, MariaDB, MySQL, SQLite, Elasticsearch, Pinecone
- **AI / GenAI** : LLMs, RAG, LangChain, LangGraph, NLP, Hugging Face, OpenAI API, Gemini API
- **Healthcare** : QT, VTK, ITK, Medical Imaging
- **DevOps & Tools** : Git, GitHub, Docker, Linux, Azure DevOps, CI/CD, Kibana, PgAdmin, Power BI

PUBLICATIONS

- 1. **Mental Health Risk Detection via Explainable AI (XAI): A Multi-Model Benchmark with Contextual Embeddings**, *International Conference on Artificial Intelligence Trends and Pattern Recognition (ICAITPR 2026)* - Under Review
- 2. **AI-Based Analysis of Hospital Bed Utilization Patterns for Patient Readmission Prediction**, *International Conference on Smart AI and Health Innovation Through Tamil for Holistic Advancement (SAHITHYA 2025), NITTTR Chennai (ISBN Conference Proceedings)* - Accepted
- 3. **Design of PON for Hospital Building Management: Enhancing Data Transmission Using Free Space Optics (FSO)**, *International Conference on Human-Centric Smart Computing (ICHCSC 2025), Springer Nature - Proceedings Accepted*

EDUCATION

Bachelor of Engineering in ECE (Hons.) Artificial Intelligence Nov 2021 – May 2025 | Tamil Nadu, India
IFET College of Engineering, Anna University
CGPA : 8.71/10 | **Rank Holder** | First Class with Distinction | Associate Secretary

ACHIEVEMENTS & AWARDS

- **Rank Holder Award** – Top Academic Rank
- **Academic Excellence Award** – Outstanding Academic Performance
- **Best Student Award** – Academic Excellence & Leadership
- **Best Innovation Project Award** – Project Expo 2025
- **Best Presentation Award** – Technical Talk Competition
- **Runner-up** – CEMP Project Expo 2022
- **LinkedIn Top Voice Badge (3×)** – Technical Knowledge Sharing
- **C-DAC Performance Recognition** – Contributions to Projects

COURSES & CERTIFICATIONS

- **Data Analytics** Professional Certificate | Google
- **Python** for Data Science | IBM
- **Generative AI** (LinkedIn Learning) | Microsoft
- **Prompt Engineering** (Cognitive Class) | IBM
- **Software Engineering** Architecture | Accenture
- **Power BI** Analyst | Microsoft

INTERESTS

- Tech Meetups
- Building Applications
- Healthcare & AI Research
- Hackathons